# Reasoning LLMs

## Module 1 — What they are, why now

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## What do we mean by "reasoning"?

**Recall** — look something up:

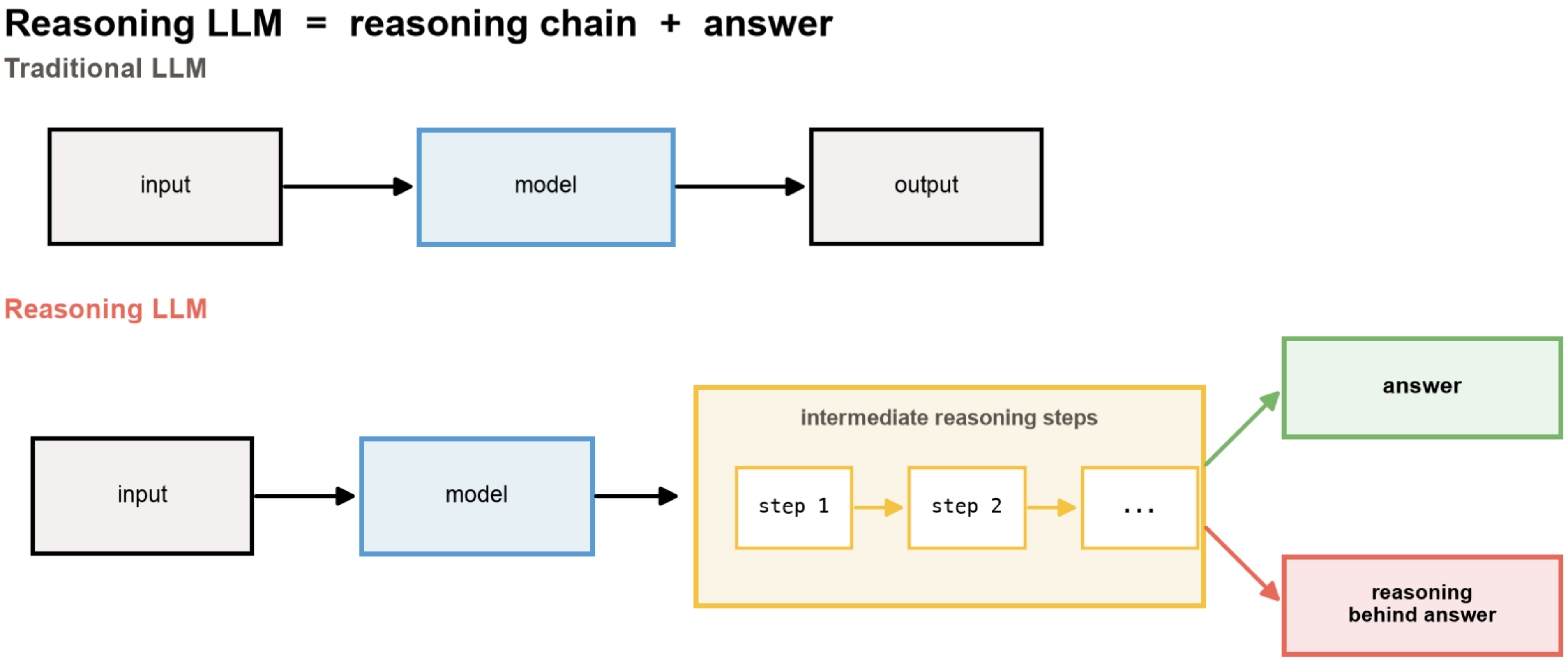
> "What's the course code for Stanford's Transformers class?" → CME 295

**Reasoning** — work it out:

> "A bear was born in 2020. How old in 2025?" → 2025 – 2020 = **5**

Reasoning = solving a problem through a **multi-step process**, not a single lookup.

# What is a reasoning LLM?

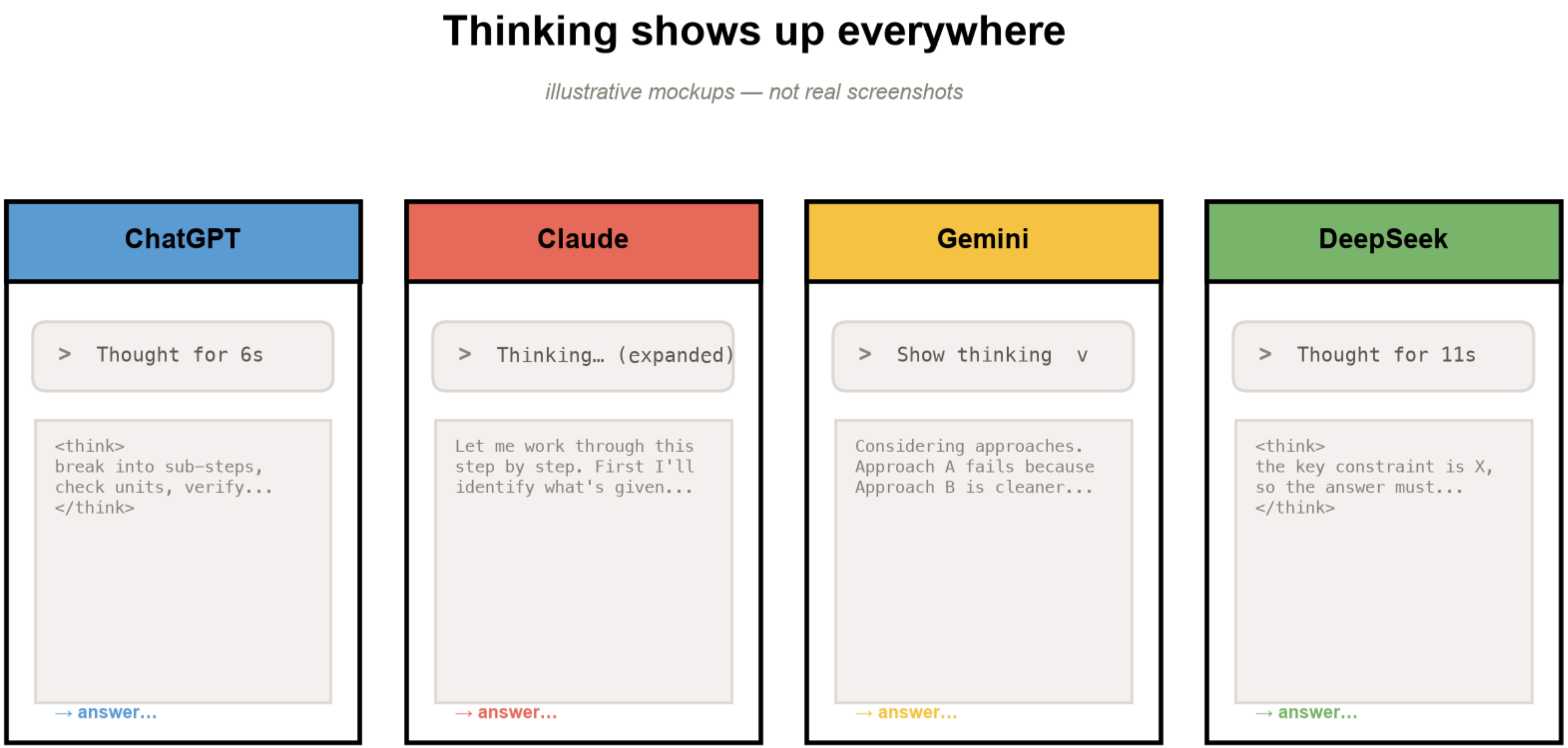



*Output = reasoning chain + answer.*



The output contract changes: **Output = reasoning chain + answer**

# You've already seen these everywhere



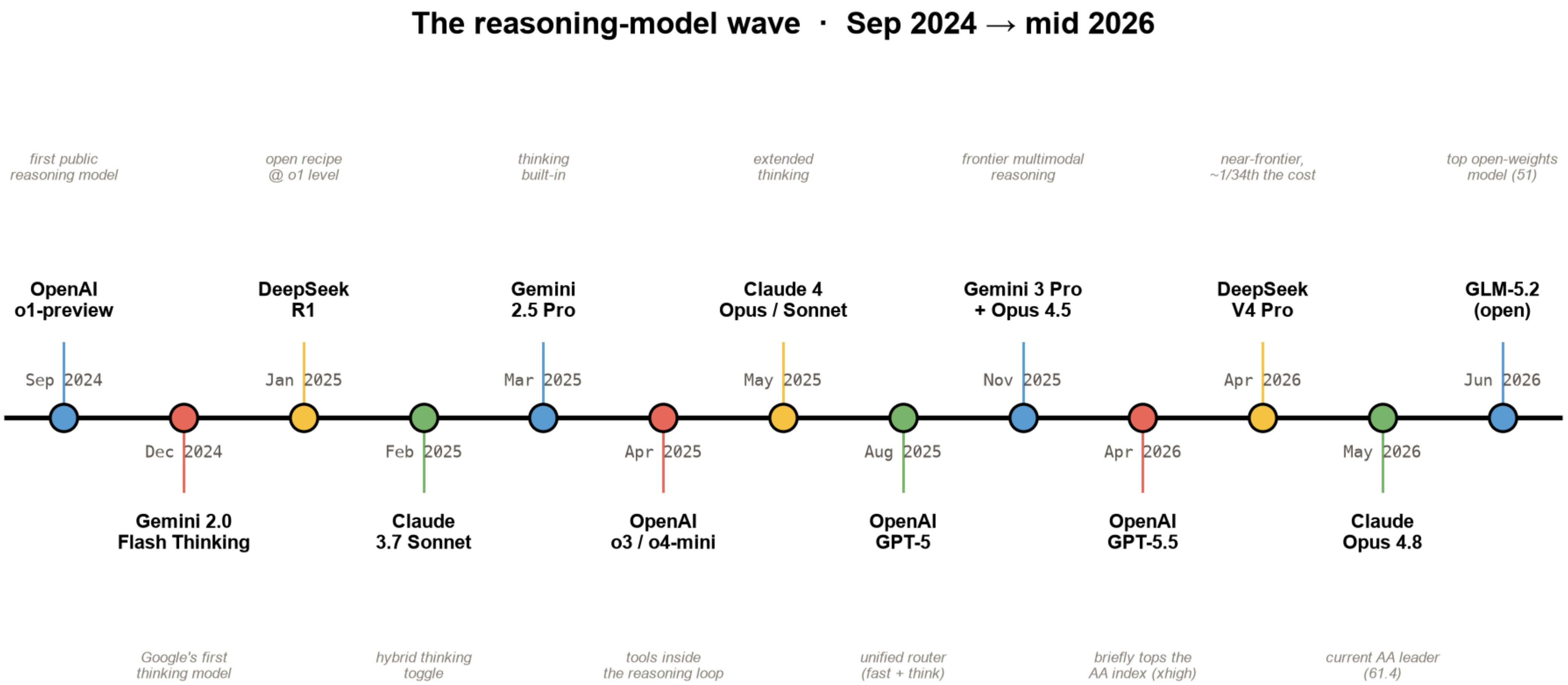
## A catch: you pay for the thoughts

Reasoning tokens are **billed as output tokens** — on every major API.

- UIs show a *summarized* thought, not the raw chain (barely-readable, and exposed chains could train rivals)
- So there's a real incentive to get **the most reasoning per token**

> More on controlling the thinking budget later.

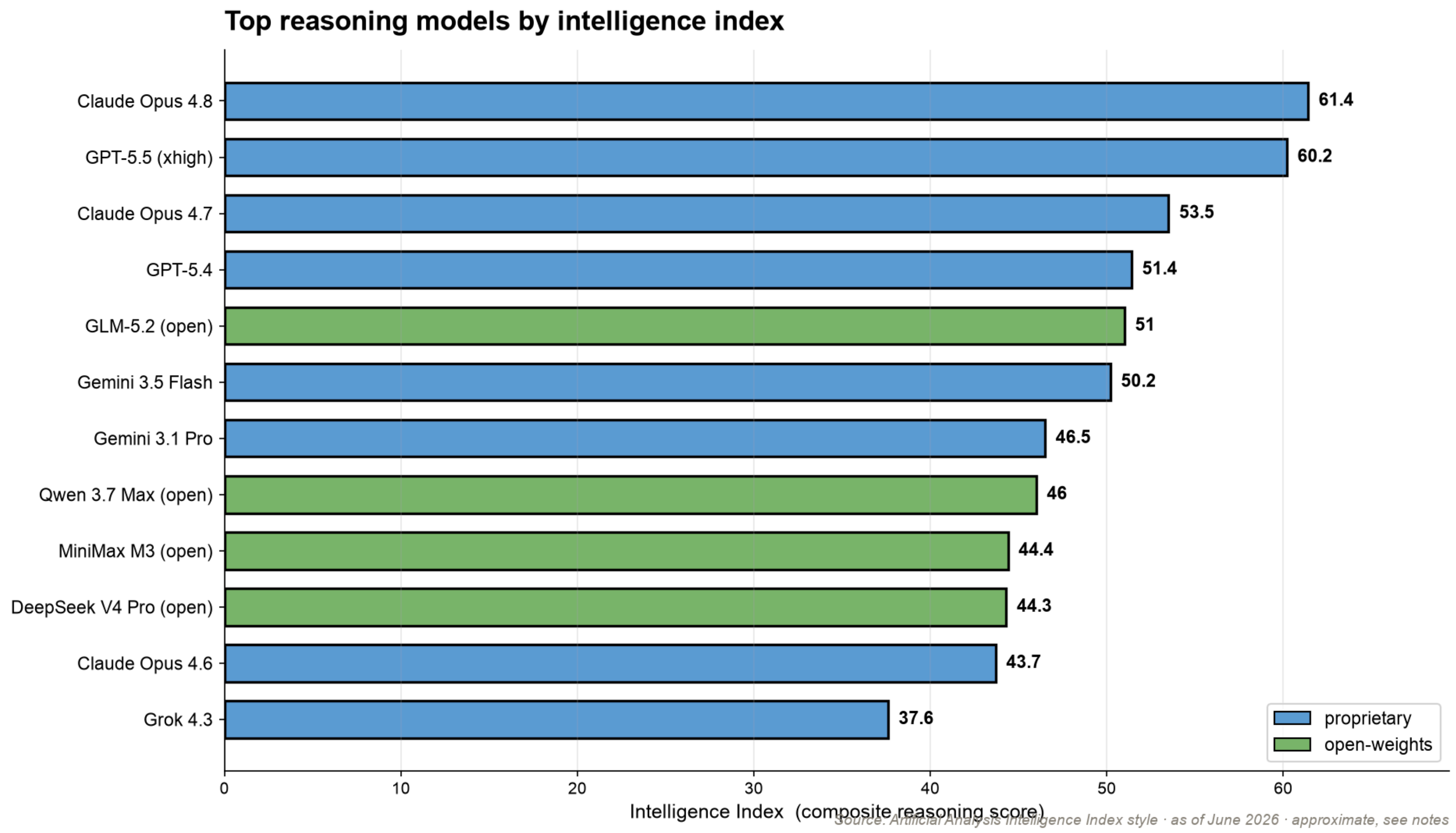
# How we got here






o1-preview kicked it off (Sep 2024). DeepSeek R1 (Jan 2025) made the recipe

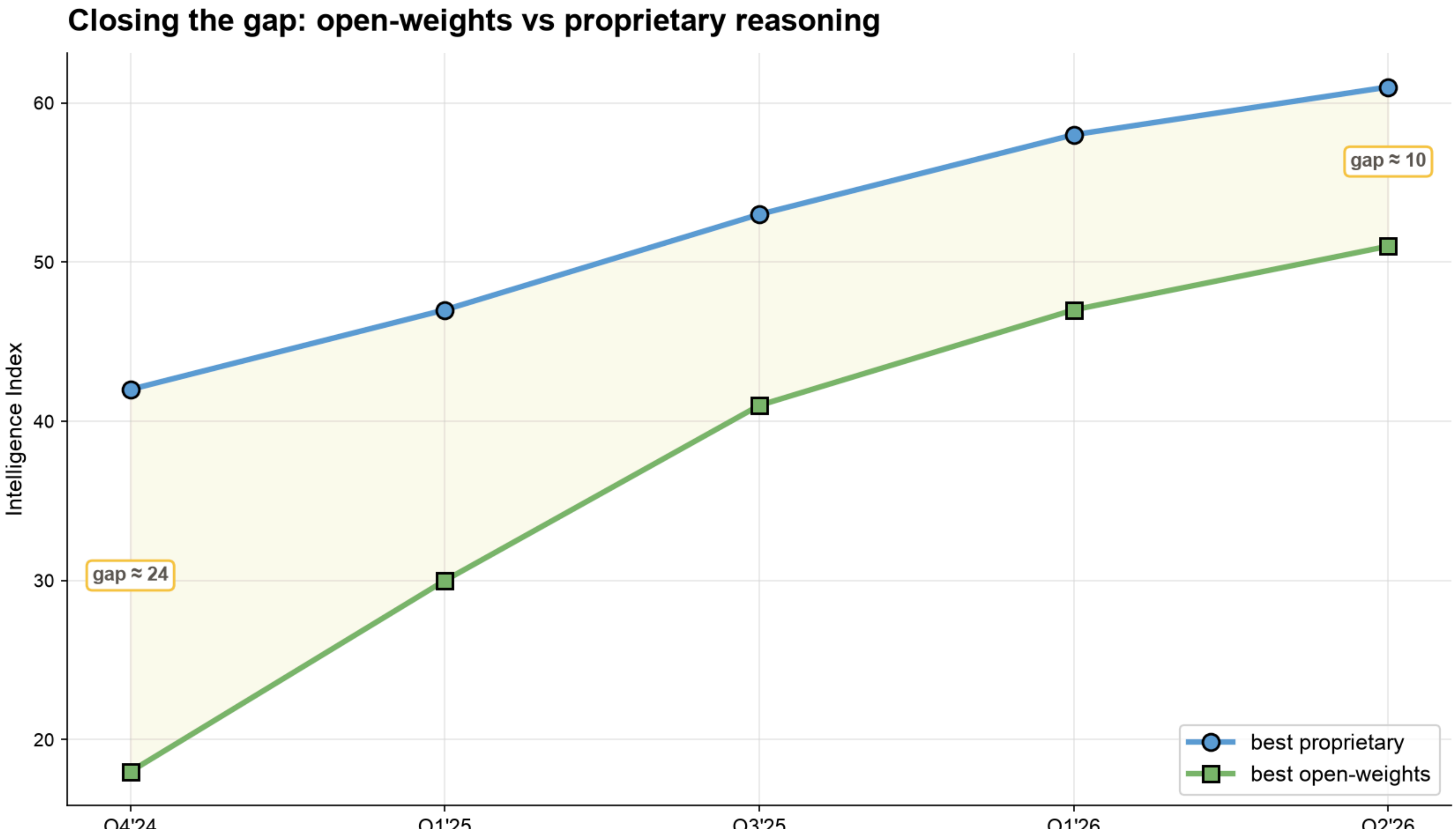
# Where the models rank today






Demand concentrates on a handful of top labs. *(Artificial Analysis, ~June 2026.)*

# Closing the gap

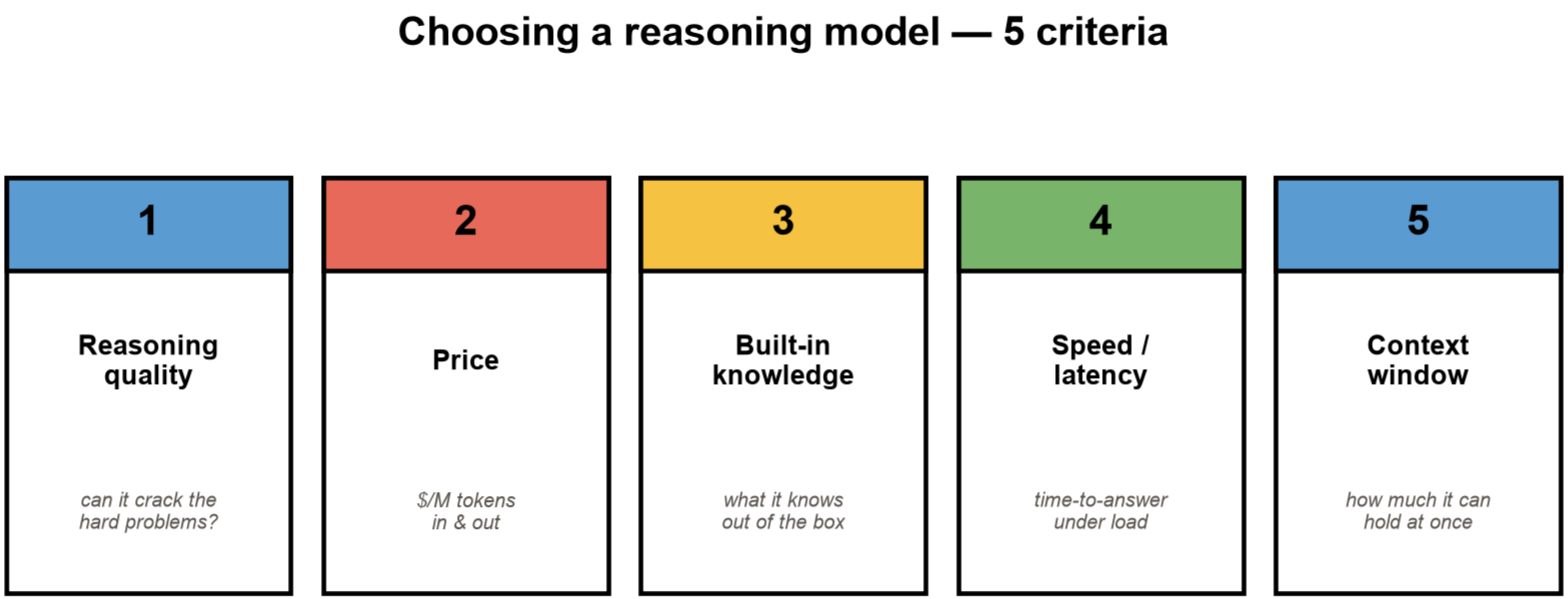


## Why reach for a reasoning model?

It earns its extra tokens when the task is a **derivation**, not a lookup:

- Decomposes hard, multi-constraint problems
- Hallucinates less on problems it can *check itself* on
- Gives you the **reasoning**, not just the verdict

# What actually matters when you pick one




> Reasoning quality · price · built-in knowledge · speed · context window.

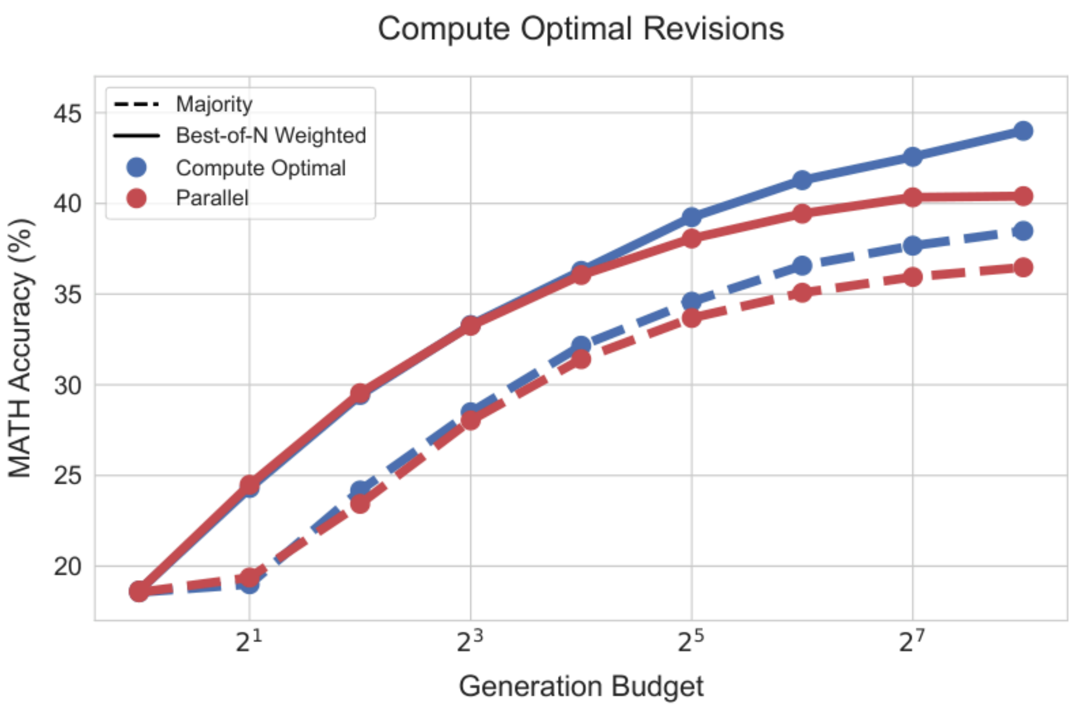
## Price is the other axis

| **Model (~June 2026)** | **$/M out** | |
|---|---|---|
| GPT-5.5 | ~$30 | frontier |
| Claude Opus 4.8 | ~$25 | frontier |
| Gemini 3.1 Pro | ~$12 | |
| **DeepSeek V4 Pro** | **~$0.87** | near-frontier, **~34× cheaper** |

> The expensive model isn't always the right call. *(Approx; we test this in notebook 08.)*

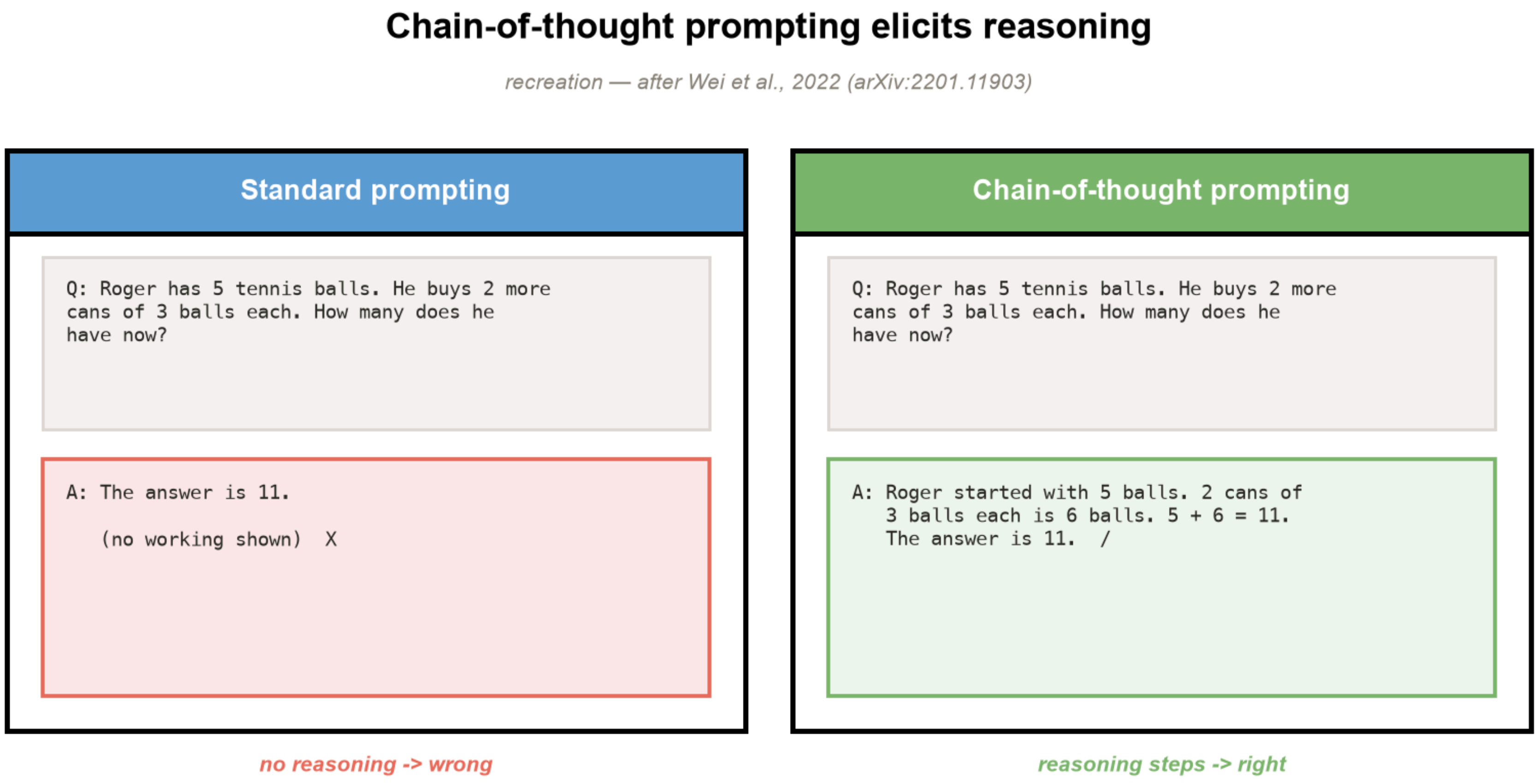
# Why does thinking longer help?

**Iteratively Revising**




> Every generated token is another forward pass — reasoning tokens **buy compute** before the model commits.

# The simplest version of the trick




> "Let's think step by step." Chain-of-thought, scaled up, *is* the core idea. *(after Wei et al., 2022)*

## Two reasons a reasoning chain helps

1. **Decomposition** — break a problem the model has never seen into sub-problems it *has*.
2. **More compute** — more tokens before the answer = more computation spent on it.

> Extended chains and self-correction ("wait, that's wrong…") **emerge from RL**, not just from prompting.

# When to use one — and when not to

## When to reach for a reasoning model

| GOOD FIT | POOR FIT |
|---|---|
| + Complex, single-shot problems | - Speed / latency-critical tasks |
| + Fewer hallucinations needed | - Cheap bulk extraction |
| + Diagnosis-style reasoning | - High-volume summarization |
| + Explanations & step-by-step | - Simple lookups & formatting |
| + Hard problems other LLMs fail | - Tight per-call cost budgets |



> Great for hard, single-shot problems. Wasteful for cheap, high-volume extraction

## How to prompt a reasoning model

- Keep the prompt **short and direct** — don't hand-write the chain of thought for it
- Frame it as an **expert generalist**; state the goal, not the steps
- Give it **clear success/failure criteria** for what a good answer looks like
- Tune **effort/budget** to the task instead of always maxing it

# You're not prompting a model — you're prompting an agent

## Prompt it like an agent, not a model

| | |
|---|---|
| `Goal` | What you ultimately want accomplished. |
| `Return format` | Exact structure / schema of the output. |
| `Warnings` | Edge cases, what to avoid, constraints. |
| `Context` | Background, data, prior state. |
| `Tools` | web search · code interpreter · retrieval |
| `Actions` | Steps it may take to reach the goal. |

*You're prompting an agent now.*

## The effort knobs

- **OpenAI:** `reasoning.effort = none / low / medium / high / xhigh`
- **Anthropic:** `thinking.budget_tokens = N` (or adaptive)
- **Google:** thinking config on the Gemini models

> Same model, dial the depth to the task.

## Limitations (keep them honest)

- 5–50× more tokens → **cost & latency**
- Early/open models: **language mixing**, sensitivity to few-shot
- The visible "thought" is a summary — and **may not be faithful**
  *(Anthropic: "reasoning models don't always say what they think")*

## Live demo →

`notebooks/01_foundations_reasoning_and_cot.ipynb`

A tiny model learns that **a scratchpad beats answering directly** — the whole idea, measured, on your laptop.

## Recap

- Reasoning ≠ retrieval — **output = reasoning + answer**
- You can *see* it (the "Thinking…" tag) and you *pay* for it
- Trade tokens for accuracy — only when the task needs it
- Next: how DeepSeek **trained** one, then we **rebuild** it in code